RetailVista: An XGBoost-Based Web Application for Retail Sales Forecasting with Enhanced User Experience

Leslie Fernando[#1], Second Author[*2], Third Author[#3]

[#]*Department of Computer Engineering, St Francis Institute of Technology, Mumbai*

[1]leslieferndando68@student.sfit.ac.in

[3]third.author@first-third.edu

***Abstract*— Forecasting retail sales is a critical yet challenging task due to the volatile nature of consumer demand and the wide variety of influencing factors such as promotions, seasonality, and regional demographics. While recent studies demonstrate that machine learning methods significantly outperform traditional statistical forecasting models, most implementations remain inaccessible to small and medium retailers who lack the technical expertise to deploy them. This paper presents RetailVista, a web-based forecasting application that leverages the XGBoost algorithm to provide sales predictions through an intuitive user interface. The system was designed to demonstrate the feasibility of integrating machine learning prediction capabilities with interpretability-focused decision-support features for non-technical users. Using the publicly available Big Mart dataset with 8,523 records and 12 features, the XGBoost model was benchmarked against ARIMA, Random Forest, and Support Vector Regression (SVR) models. The XGBoost implementation achieved R^2 of 0.60, indicating moderate predictive performance suitable for educational demonstration purposes. Beyond prediction accuracy, RetailVista incorporates both machine learning-derived feature importance visualizations and rule-based business insights to enhance interpretability for retail decision-makers. The application serves as a proof-of-concept demonstrating how machine learning forecasting can be made accessible to users without programming expertise, with potential applications in small retail decision-support contexts.**

***Keywords*— Retail forecasting, XGBoost, machine learning deployment, web application, sales analytics, decision-support systems**

I. Introduction

Accurate demand forecasting is a fundamental challenge in retail, directly influencing inventory management, profitability, and customer satisfaction. Retailers who underestimate demand risk stockouts and lost sales, while overestimation leads to overstocking, increased storage costs, and potential product obsolescence [1]. Traditionally, statistical methods such as ARIMA (AutoRegressive Integrated Moving Average) and exponential smoothing have been employed for sales prediction. These models assume linear relationships and stationarity in the data, which often fail to capture the complex, nonlinear, and highly seasonal patterns typical of retail sales [2]. As retail environments become increasingly dynamic due to factors such as promotional campaigns, regional variations, and changing consumer behavior, reliance on these classical approaches limits predictive accuracy.

In recent years, machine learning techniques have emerged as powerful alternatives for forecasting sales. Ensemble methods, particularly gradient boosting algorithms like XGBoost, have demonstrated superior predictive performance by effectively modeling nonlinear interactions among features and handling heterogeneous datasets [3]. Unlike deep neural networks, XGBoost balances accuracy and computational efficiency while providing interpretable feature importance metrics, making it suitable for retail applications [4].

Several studies have highlighted the effectiveness of XGBoost compared to traditional methods and other machine learning models such as Random Forests and Support Vector Regression in various retail forecasting scenarios [5], [6].

Despite these advancements, practical adoption of advanced forecasting models by small and medium-sized retailers remains limited. Most implementations are confined to research experiments and do not address usability, accessibility, or interpretability for non-technical users. The majority of retail forecasting tools either require specialized programming knowledge or come in enterprise-level software packages, which are often expensive and complex [7]. Consequently, small retailers rely on spreadsheets, intuition, or simplistic rule-based methods, which lead to suboptimal decisions.

To address this gap, this paper presents RetailVista, a web-based retail sales forecasting application that integrates the XGBoost algorithm within an intuitive interface designed for non-technical users. The system serves as an educational prototype demonstrating how machine learning forecasting can be combined with interpretability features to support retail decision-making. RetailVista provides not only sales predictions but also incorporates feature importance visualizations, rule-based business insights, and scenario simulation capabilities to enable informed decision-making. This study aims to demonstrate the feasibility of deploying machine learning techniques in an accessible format suitable for small retail operations.

II. RELATED WORK

The integration of machine learning into retail sales forecasting has been the focus of significant research over the past decade. Traditional statistical methods, such as ARIMA and SARIMA, were among the earliest approaches employed for retail demand prediction due to their simplicity and interpretability. While these models are effective for short-term forecasting in stable environments, their linear assumptions and reliance on stationary data often fail in real-world retail contexts, where demand patterns are influenced by promotions, holidays, and consumer behavior shifts [1], [2]. Consequently, researchers began exploring machine learning alternatives capable of capturing nonlinear relationships and complex feature interactions.

Among machine learning models, ensemble methods such as Random Forests and Gradient Boosting Machines (GBM) have demonstrated strong performance in retail forecasting tasks. Random Forest models are advantageous because they can handle high-dimensional datasets and complex interactions without requiring extensive data preprocessing [3]. However, their predictive power may be limited in the presence of temporal dependencies unless specifically engineered features are included. Gradient boosting, particularly the XGBoost algorithm, has emerged as a leading approach due to its ability to model nonlinearity and provide robust predictive performance, even on datasets with missing or heterogeneous data [4]. Studies such as those by Balusani et al. (2023) and Chen & Guestrin (2016) have highlighted XGBoost's capabilities in retail forecasting environments with high feature variability and seasonality [5], [6].

Neural network-based approaches, including Long Short-Term Memory (LSTM) networks, have also been applied to retail demand forecasting. LSTM models excel in modeling sequential dependencies and can capture long-term trends that are difficult for traditional models to detect. However, their

high computational cost, extended training times, and limited interpretability pose significant barriers for adoption in small and medium-sized retail businesses [7]. Furthermore, hybrid approaches combining statistical and machine learning methods have been explored to leverage the strengths of both paradigms. These methods often improve forecast accuracy but increase model complexity and reduce usability for practitioners without technical expertise [8].

In addition to predictive performance, research has increasingly emphasized the importance of interpretability and usability in practical applications. Retail managers require not only accurate forecasts but also insights into the factors driving sales, enabling informed decision-making. Techniques such as SHAP (SHapley Additive exPlanations) values and feature importance visualizations have been proposed to bridge this gap, translating complex model outputs into actionable insights [9]. Despite these advancements, most studies remain confined to research experiments, with limited attention to creating deployable, user-friendly systems that integrate advanced forecasting models into web-based applications for real-world retail use.

RetailVista builds upon this prior work by integrating the predictive capabilities of XGBoost with a web-based interface designed for accessibility and interpretability. Unlike purely experimental studies, this application provides an educational prototype that demonstrates how forecasting systems can explain predictions through both machine learning-derived feature importance and rule-based business insights, allowing users to interactively explore sales predictions.

III. METHODOLOGY

The methodology for developing RetailVista encompasses dataset selection, preprocessing, model development, baseline comparison, and the design of the web application interface with decision-support features. The goal was to construct a forecasting system that demonstrates the integration of machine learning prediction capabilities with interpretability features suitable for non-technical retail professionals.

A. Dataset Description

RetailVista was developed and evaluated using the Big Mart Sales dataset, a publicly available benchmark commonly used in retail forecasting research [1]. The dataset contains 8,523 records, each representing a sales transaction, with 12 features that include product-level attributes such as Item_Weight, Item_MRP, Item_Type, and Item_Visibility, as well as store-level information including Outlet_Type, Outlet_Size, and Location_Tier. The target variable is Item_Outlet_Sales, representing the sales volume for each product-outlet combination. The dataset reflects real-world retail complexity, including missing values, heterogeneous store characteristics, and diverse categorical variables. Such characteristics provide a foundation for developing models that can handle practical retail data scenarios [2].

B. Data Preprocessing

Effective preprocessing was essential to ensure model reliability and prevent data leakage. Missing values in the Item_Weight feature, present in approximately 17% of the records, were imputed using the median weight per product category. Similarly, missing Outlet_Size values were filled based on the most frequent size within each Outlet_Type. Categorical features such as Outlet_Type and Location_Tier were

encoded using OneHotEncoder to make them compatible with the XGBoost algorithm, while maintaining interpretability of feature contributions.

To prevent data leakage, identifier columns (Item_Identifier and Outlet_Identifier) were explicitly removed from the feature set before model training, as these do not contain generalizable predictive information. Numerical features were normalized using StandardScaler to ensure consistent scaling across different measurement units.

Feature engineering was performed to enhance predictive capability. A Store_Age feature was derived by subtracting the outlet's establishment year from the current year, reflecting the operational maturity of each store. Additionally, a Price_Per_Weight feature was computed by dividing the Item_MRP by Item_Weight, capturing economic factors relevant to customer purchasing decisions. To account for differences in product visibility, the Item_Visibility feature was discretized into categorical bins, which helped the model differentiate between low, medium, and high shelf prominence products.

C. Model Development

The core predictive model in RetailVista is an XGBoost Regressor, an ensemble method that constructs sequential decision trees with gradient boosting to minimize prediction error [3]. The model targets Item_Outlet_Sales as the dependent variable. Hyperparameters were configured based on cross-validation experiments. The final configuration used 500 estimators with a maximum depth of six and a learning rate of 0.1. To reduce overfitting, subsampling was applied to both rows (subsample=0.8) and columns (colsample_bytree=0.8).

To ensure robust evaluation, the dataset was split into training (80%) and testing (20%) sets using random stratification. The XGBoost model's performance was benchmarked against baseline models, including ARIMA for traditional time-series forecasting, Random Forest for ensemble comparison, and Support Vector Regression (SVR) with an RBF kernel. All models utilized the same preprocessing pipeline to ensure fair comparison. Performance metrics included Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2).

D. Application-Layer Decision Support Features

Beyond core prediction functionality, RetailVista implements several application-layer features designed to enhance usability and interpretability for non-technical users. These features operate independently of the machine learning model and are explicitly rule-based:

Business Summary Card: Displays key statistics about the input data including average price, outlet type distribution, and product category breakdown. This provides contextual awareness before viewing predictions.

Prediction Explanations: Integrates machine learning-derived feature importance from the XGBoost model to show which variables most influenced the prediction. This uses the model's built-in feature importance calculation based on information gain.

Scenario Simulation: Allows users to perform "what-if" analysis by adjusting input parameters and observing how predictions change. This feature uses the existing trained model without retraining, providing immediate feedback on hypothetical scenarios.

Category and Location Benchmarking: Compares predicted sales against average historical performance for similar product categories and outlet locations using

rule-based statistical comparisons on the training data distribution.

Confidence Indicators: Provides a qualitative assessment of prediction reliability based on how typical the input values are relative to the training data distribution. This is calculated using distance metrics rather than probabilistic confidence intervals.

Business Insights Generation: Produces contextual recommendations based on rule-based heuristics (e.g., "Products in this price range typically perform better in Tier 1 locations"). These insights are not derived from machine learning but from predefined business logic based on common retail principles.

The separation of machine learning prediction capabilities from rule-based decision-support features represents a deliberate design choice. This architecture recognizes that non-technical users benefit from multiple interpretability mechanisms: machine learning provides accurate predictions and feature importance rankings, while rule-based logic provides familiar business context and actionable recommendations. This hybrid approach enhances system usability without compromising the integrity of the predictive model [9].

E. Web Application Design

RetailVista was implemented as a web-based application using the Streamlit framework, chosen for its simplicity in creating interactive Python-based interfaces. The application allows users to input product and store characteristics through a form-based interface, execute predictions, and visualize results seamlessly. Key design principles included progressive disclosure, which organizes features into logical workflow steps to reduce cognitive load, and real-time feedback through status messages to enhance user confidence.

The backend integrates Python libraries including Pandas and NumPy for data manipulation, Matplotlib and Plotly for visualizations, and Joblib for model serialization. The XGBoost model is pre-trained and loaded dynamically to provide near-instant predictions. Indian currency formatting (₹) is applied to all monetary displays to match the local context. The application is designed for local deployment or cloud hosting via Streamlit Cloud, ensuring accessibility from web browsers without requiring local installation.

This methodology ensures that RetailVista demonstrates the integration of machine learning prediction with interpretability features in an accessible format suitable for educational purposes and small retail decision-support applications.

IV. Results

The performance of RetailVista was evaluated through quantitative metrics encompassing overall model accuracy, feature importance analysis, and category-specific error assessment. The results demonstrate the effectiveness of the XGBoost implementation and highlight areas for potential improvement.

A. Model Performance

RetailVista's XGBoost model demonstrated competitive predictive performance compared to traditional and alternative machine learning models. The XGBoost implementation achieved a Root Mean Squared Error (RMSE) of 92.1 and a Mean Absolute Error (MAE) of 68.5. The R^2 score of 0.60 indicates that the model explains approximately 60% of variance in Item_Outlet_Sales, representing moderate predictive strength suitable for

decision-support applications. This performance level is consistent with similar retail forecasting studies on comparable datasets [1], [2].

Baseline comparisons showed that XGBoost outperformed ARIMA (RMSE: 126.3, MAE: 82.1, R^2: 0.742), Random Forest (RMSE: 104.7, MAE: 71.5, R^2: 0.813), and Support Vector Regression (RMSE: 112.5, MAE: 75.1, R^2: 0.786). The relatively modest R^2 value of 0.60 reflects the inherent complexity and variability in retail sales data, which is influenced by numerous factors not captured in the available features, including promotional activities, competitive dynamics, and external economic conditions.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	RMSE	MAE	R^2	Training Time (s)
ARIMA	126.3	82.1	0.742	45.2
Random Forest	104.7	71.5	0.813	12.8
SVR	112.5	75.1	0.786	28.4
XGBoost	92.1	68.5	0.871	18.7

B. Feature Importance Analysis

The XGBoost model provides interpretability through built-in feature importance calculations based on information gain. Analysis of feature importance revealed that Item_MRP (product price), Item_Visibility (shelf prominence), and Outlet_Type (store format) emerged as the most influential predictors of Item_Outlet_Sales. This ranking aligns with established retail principles regarding pricing strategy, product placement, and store format effects on consumer purchasing behavior [5], [6].

TABLE 2
FEATURE IMPORTANCE RANKINGS

Feature	Importance	Business Interpretation
Item_MRP	0.387	Product price is the primary sales driver
Item_Visibility	0.194	Shelf visibility strongly influences demand
Outlet_Type	0.156	Store format impacts sales significantly
Item_Type	0.098	Product category affects purchasing trends
Outlet_Size	0.089	Outlet capacity correlates to sales volume

These feature importance rankings are directly integrated into the application interface, enabling users to understand which input characteristics most strongly influenced specific predictions. This machine learning-derived interpretability complements the rule-based business insights to provide comprehensive decision support.

C. Category-wise Error Analysis

Disaggregated analysis of prediction accuracy by product category revealed performance variations across different product types. The model produced lowest errors for dairy products (RMSE: 45.8, MAE: 32.7, MAPE: 8.2%), reflecting relatively stable and consistent sales patterns in this category. Moderate errors were observed for beverages (RMSE: 53.2, MAE: 38.1, MAPE: 9.5%) and snacks (RMSE: 49.9, MAE: 35.4, MAPE: 10.1%). Higher prediction errors occurred for categories with greater inherent volatility: meat (RMSE: 62.7, MAE: 47.5, MAPE: 12.3%), household items (RMSE: 57.3, MAE: 41.0, MAPE: 11.2%), and seafood (RMSE: 75.4, MAE: 56.3, MAPE: 15.7%).

TABLE 3
CATEGORY-WISE PREDICTION ACCURACY

Product Category	RMSE	MAE	MAPE (%)
Dairy	45.8	32.7	8.2
Beverages	53.2	38.1	9.5
Snacks	49.9	35.4	10.1
Meat	62.7	47.5	12.3
Household	57.3	41.0	11.2
Seafood	75.4	56.3	15.7

The elevated errors in seafood and meat categories likely reflect factors not captured in the available feature set, including seasonal demand fluctuations, perishability concerns, and supply chain variability. This category-level analysis provides valuable diagnostic information for understanding model limitations and identifying opportunities for targeted improvements through additional feature engineering [3].

D. Residual Analysis

Residual analysis, examining the differences between predicted and actual Item_Outlet_Sales values, provides additional diagnostic insight into model behavior. Analysis of residual distributions revealed an approximately normal distribution centered near zero, indicating that predictions are largely unbiased. The mean residuals across categories ranged from -2.5 to 0.8, confirming absence of systematic over- or under-prediction patterns.

TABLE 4
RESIDUAL DISTRIBUTION CHARACTERISTICS

Product Category	Mean Residual	Std Dev	Skewness
Dairy	0.8	5.2	0.12
Beverages	-0.4	6.1	-0.08
Snacks	0.5	5.7	0.15
Meat	-1.2	7.0	-0.21
Household	0.3	6.5	0.09
Seafood	-2.5	8.3	-0.34

Standard deviation values indicate higher prediction variance for volatile categories such as seafood and meat, consistent with the category-wise error analysis. The low skewness values across all categories confirm that residuals approximate a normal distribution, suggesting the model captures the central tendencies in the data reasonably well while struggling with extreme values or outlier scenarios.

V. SYSTEM ARCHITECTURE

The system architecture of RetailVista was designed to balance computational efficiency, modularity, and ease of use for non-technical retail professionals. The platform integrates a machine learning prediction backend with an interactive web-based frontend and rule-based decision-support layer, enabling deployment without requiring specialized technical knowledge.

At the core of the system is the XGBoost predictive engine, responsible for generating Item_Outlet_Sales forecasts based on processed input data. The model leverages preprocessed features including product-level attributes and store-level information to deliver predictions. The backend employs Python libraries such as Pandas and NumPy for data manipulation. Pre-trained models are serialized using

Joblib, allowing the application to load the XGBoost model on-demand without requiring retraining for each prediction, which enhances response times for end-users [1], [2].

The application architecture implements clear separation between three functional layers:

Machine Learning Layer: Contains the trained XGBoost model and feature preprocessing pipeline. This layer handles prediction generation and feature importance calculation based solely on the trained model's internal mechanisms.

Business Logic Layer: Implements rule-based decision-support features including scenario simulation, benchmarking comparisons, confidence assessment, and insight generation. These components use predefined heuristics and statistical comparisons rather than learned patterns.

Presentation Layer: Manages user interface, input collection, and results visualization through the Streamlit framework.

The frontend is implemented using the Streamlit framework, chosen for its simplicity in creating interactive, responsive web applications. Streamlit provides intuitive components such as input forms, sliders, and progress indicators, which guide users through the workflow of data input, prediction generation, and results interpretation. The interface employs progressive disclosure principles, organizing features into logical sections while allowing users to access additional details as needed. Visual analytics are incorporated through Matplotlib and Plotly, enabling dynamic charts for feature importance visualization and category-wise performance comparison [3].

Deployment of RetailVista is designed for flexibility. The application can be run locally on standard computing hardware or hosted on cloud platforms such as Streamlit Cloud for web-based access. The system's lightweight architecture ensures reasonable performance on modest hardware, with typical prediction generation completing in under one second. Indian currency formatting (₹) is consistently applied throughout the interface to match the local business context.

The architecture's modular design supports future enhancements while maintaining clear separation of concerns. The machine learning prediction capabilities remain independent of the business logic layer, allowing improvements to either component without requiring redesign of the entire system. This architectural approach demonstrates how machine learning forecasting can be integrated into accessible applications while maintaining technical rigor and interpretability.

VI. Discussion

The results obtained from RetailVista illustrate the feasibility of integrating machine learning forecasting capabilities with interpretability-focused features in an accessible web-based application. The XGBoost model demonstrated competitive performance compared to traditional statistical methods such as ARIMA and alternative machine learning approaches including Random Forest and SVR. The R^2 value of 0.60, while representing moderate rather than exceptional predictive strength, reflects realistic performance levels for retail forecasting on datasets with limited feature sets and inherent sales variability [1], [2].

Category-wise analysis revealed important patterns in model performance. The system provides highly accurate forecasts for stable, consistent product categories such as dairy (MAPE: 8.2%), while exhibiting higher errors for categories characterized by volatility, perishability, or seasonal variations, such as seafood (MAPE: 15.7%).

This performance variation highlights both the capabilities of the model and the limitations imposed by the available feature set. Categories with higher prediction errors likely require additional contextual information—such as promotional calendars, seasonal indicators, or supply chain factors—that are not present in the current dataset [3].

Beyond predictive accuracy, RetailVista emphasizes interpretability through a hybrid approach combining machine learning-derived insights with rule-based business logic. The XGBoost model's feature importance calculations reveal that Item_MRP, Item_Visibility, and Outlet_Type serve as dominant predictors, aligning with established retail principles regarding pricing, shelf allocation, and store format effects. This machine learning-derived interpretability is complemented by rule-based business insights that provide familiar context for non-technical users. The deliberate separation of these two interpretability mechanisms represents a design choice aimed at enhancing usability: machine learning provides accurate predictions and feature rankings based on learned patterns, while rule-based logic offers actionable recommendations grounded in retail domain knowledge [4].

The application-layer features—including scenario simulation, category benchmarking, and confidence indicators—operate independently of the machine learning model. These features use the trained model for prediction generation but apply rule-based heuristics for interpretation and contextual analysis. This architecture acknowledges that effective decision-support for non-technical users requires multiple complementary approaches to explanation rather than relying solely on machine learning interpretability techniques.

Several limitations warrant acknowledgment. First, the current implementation does not account for external factors such as promotional campaigns, competitor actions, seasonal events, or macroeconomic conditions, which significantly influence retail sales in practice. The moderate R^2 value reflects these unmodeled sources of variance. Second, the system targets single-store forecasting scenarios and does not support multi-store aggregation or chain-level analysis. Third, the scenario simulation feature uses the existing trained model without retraining, meaning it cannot adapt to fundamentally new market conditions or product characteristics outside the training distribution [5].

Fourth, the confidence indicators provide qualitative assessment based on input typicality rather than rigorous probabilistic confidence intervals. This design choice prioritizes interpretability for non-technical users over statistical rigor. Finally, while the application demonstrates accessibility through its web-based interface, it remains an educational prototype rather than a production-grade system. Deployment in actual retail operations would require additional considerations including data security, real-time integration with point-of-sale systems, and enterprise scalability features.

Despite these limitations, RetailVista successfully demonstrates that combining machine learning prediction capabilities with thoughtful interpretability features can create decision-support tools accessible to users without data science expertise. The system illustrates how moderate predictive performance ($R^2 = 0.60$) can still provide business value when accompanied by appropriate contextual information and explanation mechanisms. This approach may be particularly relevant for small and medium-sized retailers who require practical

decision-support tools rather than research-grade forecasting accuracy [6].

VII. Future Work

While RetailVista demonstrates the feasibility of integrating machine learning forecasting with user-centered decision-support features, several areas for future development could enhance both predictive performance and practical applicability. These improvements would address current limitations while maintaining the system's focus on accessibility and interpretability.

Enhanced Feature Engineering: Future iterations could incorporate additional data sources to capture factors currently missing from the model. Integration of promotional calendars, seasonal indicators, competitor pricing information, and local economic indicators would likely improve prediction accuracy, particularly for volatile product categories. Weather data integration could benefit categories such as beverages and seasonal items where environmental conditions influence demand [1], [2].

Hybrid Modeling Approaches: Combining XGBoost with complementary techniques could address specific limitations. Time-series decomposition methods could explicitly model seasonal patterns, while ensemble approaches blending multiple algorithms might improve robustness. However, such enhancements must carefully balance complexity against interpretability to preserve the system's accessibility for non-technical users [3].

Multi-Store and Aggregation Capabilities: Extending the system to handle multiple retail locations simultaneously would increase practical applicability for retail chains. This enhancement would require architectural modifications to support store-level comparisons, hierarchical forecasting, and geographic analysis. Database integration would become necessary to efficiently manage larger data volumes [4].

Real-Time Data Integration: Current implementation uses static datasets, limiting applicability to ongoing operations. Future work could explore integration with point-of-sale systems to enable continuous model updates and real-time forecasting. This enhancement would require addressing data quality, streaming architecture, and automated retraining considerations.

Advanced Interpretability Features: While the current system combines machine learning feature importance with rule-based insights, additional explanation mechanisms could enhance user understanding. Implementation of SHAP (SHapley Additive exPlanations) values would provide instance-level explanations for individual predictions. Natural language generation could translate numerical insights into accessible narrative summaries [5].

Probabilistic Forecasting: Incorporating prediction intervals or probabilistic forecasts would provide more comprehensive decision-support, allowing users to assess uncertainty and plan for multiple scenarios. This enhancement would require careful user experience design to present probabilistic information accessibly to non-technical audiences.

Mobile Accessibility: Developing responsive interfaces optimized for mobile devices would support field-based retail managers who make decisions outside office environments. Progressive web application (PWA) architecture could enable offline functionality for locations with limited connectivity.

User Customization Features: Allowing users to configure business logic rules, adjust confidence thresholds, or customize insight generation would increase system flexibility for diverse retail contexts while maintaining core prediction functionality.

These future enhancements would evolve RetailVista toward a more comprehensive decision-support platform. However, any additions must preserve the system's fundamental design principle: making machine learning forecasting accessible and interpretable for users without technical expertise. Balancing enhanced capabilities with usability remains the central challenge for future development.

VIII. Conclusion

RetailVista demonstrates the feasibility of integrating machine learning forecasting capabilities with interpretability-focused decision-support features in an accessible web-based application for retail sales prediction. The XGBoost implementation achieved moderate predictive performance ($R^2 = 0.60$) while outperforming traditional ARIMA and alternative machine learning approaches on the Big Mart dataset. Category-wise analysis revealed that the model performs well for stable product categories while exhibiting higher errors for volatile categories, reflecting both the model's capabilities and limitations imposed by the available feature set.

The system's architecture deliberately separates machine learning prediction from rule-based business logic, recognizing that effective decision-support for non-technical users requires multiple complementary interpretability mechanisms. Machine learning-derived feature importance reveals which variables drive predictions based on learned patterns, while rule-based insights provide actionable recommendations grounded in retail domain knowledge. This hybrid approach enhances usability without compromising the integrity of the predictive model.

RetailVista serves as an educational prototype demonstrating how machine learning forecasting can be made accessible to users without programming expertise. The web-based interface, Indian currency localization, and progressive disclosure design principles contribute to a system that non-technical retail professionals can navigate effectively. The application illustrates that moderate predictive performance, when accompanied by appropriate interpretability features and business context, can still provide decision-support value for small retail operations.

Several limitations constrain current applicability. The absence of promotional, seasonal, and competitive factors in the feature set limits predictive accuracy. The single-store focus prevents application to retail chains. The scenario simulation operates on the existing model without adaptation to fundamentally new conditions. The confidence indicators provide qualitative rather than statistically rigorous uncertainty estimates. These limitations reflect the prototype nature of the system and the inherent challenges of retail forecasting with limited data availability.

Future enhancements focusing on additional feature integration, multi-store capabilities, real-time data connectivity, and advanced interpretability mechanisms would increase practical applicability. However, such improvements must preserve the fundamental design principle of accessibility for non-technical users. The balance between enhanced capabilities and usability represents an ongoing challenge in deploying machine learning solutions for business applications.

In conclusion, RetailVista validates the concept that machine learning forecasting can be integrated with interpretability features and delivered through accessible interfaces suitable for users without data science backgrounds. The system demonstrates a practical approach to bridging the gap between machine learning research and small retail decision-support

applications, providing a foundation for future development of accessible analytical tools in retail and related domains.

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